

6. GLYCEMIC GOALS, HYPOGLYCEMIA, AND HYPERGLYCEMIC CRISES

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6. Glycemic Goals, Hypoglycemia, and Hyperglycemic Crises: Standards of Care in Diabetes—2026

American Diabetes Association Professional Practice Committee for Diabetes*

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The American Diabetes Association (ADA) “Standards of Care in Diabetes” includes the ADA’s current clinical practice recommendations and is intended to provide the components of diabetes care, general treatment goals and guidelines, and tools to evaluate quality of care. Members of the ADA Professional Practice Committee for Diabetes, an interprofessional expert committee, are responsible for updating the Standards of Care annually, or more frequently as warranted. For a detailed description of ADA standards, statements, and reports, as well as the evidence-grading system for ADA’s clinical practice recommendations and a full list of Professional Practice Committee members, please refer to Introduction and Methodology. Readers who wish to comment on the Standards of Care are invited to do so at professional.diabetes.org/SOC.

ASSESSMENT OF GLYCEMIC STATUS

Glycemic status is assessed by A1C measurement, blood glucose monitoring (BGM) by capillary (finger-stick) devices, and different continuous glucose monitoring (CGM) metrics such as time in range (TIR), time below range (TBR), time above range (TAR), glucose management indicator (GMI), coefficient of variation, and mean glucose. Clinical trials of interventions that lower A1C have demonstrated the benefits of improved glycemia with respect to long-term diabetes complications. Glucose monitoring via CGM or BGM (discussed in detail in section 7, “Diabetes Technology”) is useful for diabetes self-management, can provide nuanced information on glucose responses to meals, physical activity, and medication changes, and is particularly useful for optimal medication management and safety in individuals taking insulin. CGM serves an increasingly important role in optimizing the effectiveness and safety of treatment in many people with type 1 diabetes, type 2 diabetes, diabetes in pregnancy, and other forms of diabetes (e.g., cystic fibrosis–related diabetes). Individuals on a variety of insulin treatment plans benefit from CGM with improved glucose levels, decreased hypoglycemia, and enhanced self-efficacy (section 7, “Diabetes Technology”) (1).

Glycemic Assessment Recommendations

6.1 Assess glycemic status by A1C A and/or continuous glucose monitoring (CGM) metrics such as time in range, time above range, and time below range.

B Fructosamine or CGM can be used for glycemic monitoring when an alternative to A1C is required. B

6.2 Assess glycemic status at least two times a year, and more frequently (e.g., every 3 months) for individuals not meeting glycemic goals or with recent treatment changes, frequent or severe hypoglycemia or hyperglycemia, or changes in health status, or during periods of rapid growth and development in children and adolescents. E

*A complete list of members of the American Diabetes Association Professional Practice Committee for Diabetes can be found at <https://doi.org/10.2337/dc26-SINT>.

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diabetesjournals.org/care # Glycemic Goals, Hypoglycemia, and Hyperglycemic Crises

Glycemic Assessment by A1C

The A1C test is the primary tool for assessing glycemic status in both clinical practice and clinical trials, and it is strongly linked to diabetes complications (2–4). A1C reflects average glycemia over approximately 2–3 months. The performance of laboratory tests for A1C is generally excellent for National Glycohemoglobin Standardization Program (NGSP)-certified assays (ngsp.org). Thus, A1C testing should be performed routinely in all people with diabetes at initial assessment and as part of continuing care. Measurement approximately every 3 months determines whether glycemic goals have been reached and maintained. Adults with type 1 or type 2 diabetes who have achieved and are maintaining glucose levels within their goal range may only need A1C testing twice a year. Individuals with less stable glucose levels, those with intensive care plans, or those not meeting their treatment goals may require more frequent testing, typically every 3 months, with additional assessments as needed. Point-of-care A1C testing can offer timely opportunities for treatment adjustments during appointments with health care professionals, although point-of-care tests may be less accurate than laboratory A1C assays (5).

The A1C test is an indirect measure of average glycemia. Factors that affect hemoglobin or red blood cells may affect A1C; see section 2, “Diagnosis and Classification of Diabetes,” for details. For example, conditions that affect red blood cell turnover (e.g., hemolytic anemia and other anemias, glucose-6-phosphate dehydrogenase deficiency, recent blood transfusion, use of drugs that stimulate erythropoiesis, kidney failure, and pregnancy) can interfere with the accuracy of A1C (6). Some hemoglobin variants can interfere with some A1C assays; however, most assays in use in the U.S. are accurate in individuals who are heterozygous for the most common variants (7). A1C cannot be measured in individuals with sickle cell disease (HbSS) or other homozygous hemoglobin variants (e.g., HbEE), since these individuals lack HbA (8). In individuals with conditions that interfere with the interpretation of A1C, alternative approaches to monitoring glycemic status should be used, including self-monitoring of blood glucose, CGM, and/or the use of glycated serum protein assays (discussed below).

A1C does not provide a measure of glycemic variability, real-time glucose levels, or hypoglycemia. For individuals prone to glycemic variability, especially people with type 1 diabetes or type 2 diabetes with insulin deficiency and/or treatment with intensive insulin therapy, glycemic status is best evaluated by the combination of results from BGM or CGM and A1C. Discordant results between A1C and BGM or CGM can occur due to high glycemic variability, inaccurate BGM or CGM measurement, or inaccurate A1C due to the factors discussed above. As discussed in section 2, “Diagnosis and Classification of Diabetes,” there is controversy regarding the clinical significance of differences in A1C by self-reported race and ethnicity (9–12). There is an emerging understanding of genetic determinants that may modify the association between A1C and glucose levels (13). However, race and ethnicity are not good proxies for these genetic differences that are likely present in a small fraction of individuals of all racial groups. Therefore, race and ethnicity should not be considerations for how A1C is used clinically for glycemic monitoring. Limitations of laboratory tests and within-person variability in glucose and A1C underscore the importance of using multiple approaches to glycemic monitoring and further evaluation of discordant results in all racial or ethnic groups.

Serum Glycated Protein Assays as Alternatives to A1C

Fructosamine and glycated albumin are alternative measures of glycemia that are approved for clinical use for monitoring glycemic status in people with diabetes. Fructosamine reflects total glycated serum proteins (mostly albumin). Glycated albumin assays reflect the proportion of total albumin that is glycated. Due to the turnover rate of serum protein, fructosamine and glycated albumin reflect glycemia over the past 2–4 weeks, a shorter-term time frame than that of A1C. Fructosamine and glycated albumin are highly correlated in people with diabetes, and the performance of modern assays is typically excellent. Fructosamine and glycated albumin have been linked to long-term complications in epidemiologic cohort studies (14–18). However, there have been few clinical trials, and the evidence base supporting the use of these biomarkers to monitor glycemic status is much weaker than that for A1C. In people with diabetes who have conditions where the interpretation of A1C may be problematic or when A1C cannot be measured (e.g., homozygous hemoglobin variants), fructosamine or glycated albumin may be useful alternatives to monitor glycemic status (8).

Correlation Between A1C and Blood Glucose Monitoring and Continuous Glucose Monitoring

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Glycemic Goals, Hypoglycemia, and Hyperglycemic Crises

serve as values to guide changes in therapy. For older adults with complex or intermediate health using CGM, the recommended percent time spent in goal range of 70–180 mg/dL is 50% (or 12 h per day) and the recommended time spent in hypoglycemia of less than 70 mg/dL should not be more than 1%, or 15 min per day, to minimize hypoglycemia risk. In this population, more permissive hyperglycemia is allowed (up to 50% of the time in 24 h) if required to prevent hypoglycemia.

GLYCEMIC GOALS

Recommendations

- 6.3a An A1C goal of $\leq 7\%$ (≤ 53 mmol/mol) is appropriate for many nonpregnant adults without severe hypoglycemia or hypoglycemia affecting health or quality of life. A - 6.3b A goal time in range of $>70\%$ in people using CGM is appropriate for many nonpregnant adults. B - 6.3c A goal percent time ≤ 70 mg/dL (≤ 3.9 mmol/L) of $\leq 4\%$ (or $\leq 1\%$ for older adults) and a goal percent time ≤ 54 mg/dL (≤ 3.0 mmol/L) of $\leq 1\%$ are recommended in people using CGM to prevent hypoglycemia. Deintensify or modify therapy if these goals are not met. B - 6.4 Lower A1C goals (e.g., $\leq 6.5\%$ [≤ 48 mmol/mol]) may be appropriate for individuals with diabetes with good health and function and low treatment risks (e.g., hypoglycemia) and burdens (see Fig. 6.1). B - 6.5 Less stringent glycemic goals may be appropriate for individuals with significant cognitive and/or functional limitations, frailty, or severe comorbidities or where the harms of treatment, including hypoglycemia, are greater than the benefits. B - 6.6 Deintensify hypoglycemia-causing medications (insulin, sulfonylureas, or meglitinides), or switch to a medication class with lower hypoglycemia risk, for individuals who are at high risk for hypoglycemia, within individualized glycemic goals. B - 6.7 Deintensify diabetes medications for individuals for whom the harms and/or burdens of treatment may be greater than the benefits, within individualized glycemic goals. B - 6.8 Reassess glycemic goals based on the individualized criteria shown in Fig. 6.1. E

appropriate for individuals with significant functional and cognitive impairments. For more details regarding glycemic goals in older adults, please refer to section 13, “Older Adults.” For glycemic goals in children and adolescents, please refer to section 14, “Children and Adolescents.” For glycemic goals during pregnancy, please refer to section 15, “Management of Diabetes in Pregnancy.” Health care professionals should engage in shared decision-making with the individual (as well as with family members and care partners) to establish treatment goals and should adjust goals to improve safety and medication-taking behavior particularly in the setting of changing health status. Setting specific glycemic (and other) treatment goals during clinical consultations has been demonstrated to improve glycemic outcomes for individuals with diabetes.

Glucose Lowering and Microvascular Complications

The Diabetes Control and Complications Trial (DCCT), a prospective randomized controlled trial of intensive (mean A1C $\sim 7\%$ [~ 53 mmol/mol]) versus standard (mean A1C $\sim 9\%$ [~ 75 mmol/mol]) glycemic management in people with type 1 diabetes, showed definitively that better glycemic status is associated with 50–76% reductions in rates of development and progression of microvascular complications (retinopathy, neuropathy, and chronic kidney disease). Follow-up of the DCCT cohorts in the Epidemiology of Diabetes Interventions and Complications (EDIC) study demonstrated persistence of these microvascular benefits over two decades despite the fact that the glycemic separation between the treatment groups diminished and disappeared during follow-up.

Table 6.3—Summary of glycemic goals for many nonpregnant adults with diabetes

A1C $<7.0\%$ (<53 mmol/mol)† ----- ----- Preprandial capillary plasma glucose 80–130 mg/dL* (4.4–7.2 mmol/L) Peak postprandial capillary plasma glucose‡ <180 mg/dL* (<10.0 mmol/L)

Preterax and Diamicon MR Controlled Evaluation [ADVANCE], and Veterans Affairs Diabetes Trial [VADT]) were conducted to examine the effects of near normalization of blood glucose on cardiovascular outcomes. The ADVANCE and VADT trials found modest reduction in nephropathy with intensive glycemic management; ACCORD was stopped after a median of 3.5 years due to higher mortality in the intervention arm (51–55). Importantly, these landmark studies were conducted prior to the approval of GLP-1 RAs, dual GIP and GLP-1 RA, and SGLT2 inhibitors, and intensive glucose lowering was achieved predominantly through greater use of insulin and other medications. Findings from these studies, including the concerning increase in mortality in the intensive treatment arm of ACCORD, suggest caution is needed in treating diabetes to near-normal A1C goals in people with long-standing type 2 diabetes using medications with a high risk for hypoglycemia and with other adverse effects and no additional cardio-kidney-metabolic benefits.

other CVD risk factors, such as hypertension or hypercholesterolemia. likely driven by differences in study treatments and populations. However, a meta-analysis of individual participant data from UKPDS, ACCORD, ADVANCE, and VADT demonstrated a significant reduction in myocardial infarctions and major CVD events but no difference in stroke, heart failure, or mortality between intensive and less intensive glycemic management (57). Additionally, longer-term epidemiological follow-up of these studies also showed a pattern of CVD benefit (58–60). In the post-DCCT follow-up of the EDIC cohort, participants previously randomized to the intensive treatment arm had a significant 57% reduction in the risk of the composite outcome of nonfatal myocardial infarction, nonfatal stroke, or cardiovascular death compared with those previously randomized to the standard treatment arm (58). The benefit of intensive glycemic management in the EDIC cohort of people with type 1 diabetes has been shown to persist for 30 years and was associated with a 32% reduction in CVD events (59).

Glucose Lowering and Cardiovascular Disease Outcomes

The DCCT in individuals with type 1 diabetes and the UKPDS, ACCORD, ADVANCE, and VADT studies in type 2 diabetes all sought to address whether intensive glycemic management reduced cardiovascular disease (CVD) events (43,51,52,54). ACCORD, ADVANCE, and VADT were conducted in relatively older participants with a longer duration of diabetes (mean duration 8–11 years) and either CVD or multiple cardiovascular risk factors. Details of these studies are reviewed extensively in the joint ADA position statement “Intensive Glycemic Control and the Prevention of Cardiovascular Events: Implications of the ACCORD, ADVANCE, and VA Diabetes Trials” (56). No significant reduction in composite CVD events was demonstrated at the end of the intervention in any of these studies, and ACCORD was stopped prematurely at 3.5 years because of an increase in total mortality, particularly sudden CVD deaths. Serious concerns with the intensive glycemic treatment plan used in ACCORD included the rapid escalation of therapies, the early use of large doses of insulin, substantial weight gain, and frequent hypoglycemia. Blood glucose has subsequently been shown to be a weaker independent CVD risk factor compared with other CVD risk factors.

As discussed further below, severe hypoglycemia is strongly associated with cardiovascular events and mortality (64). Therefore, health care professionals should be vigilant in preventing hypoglycemia and should not aggressively attempt to achieve near-normal A1C levels in people in whom such goals cannot be safely and reasonably achieved. As discussed in section 9, “Pharmacologic Approaches to Glycemic Treatment,” addition of SGLT2 inhibitors and/or GLP-1–based therapies that have demonstrated CVD and kidney benefits is recommended in individuals with established CVD, chronic kidney disease, and heart failure. As outlined in more detail in section 9, “Pharmacologic Approaches to Glycemic Treatment,” and section 10, “Cardiovascular Disease and Risk Management,” the cardiovascular benefits of SGLT2 inhibitors or GLP-1 RAs are not contingent upon A1C lowering; therefore, initiation should be considered in people with type 2 diabetes and CVD, kidney disease, heart failure, and/or obesity independent of the current A1C, A1C goal, or metformin therapy. Based on these considerations, the following two strategies are offered (65):

1. If already on dual therapy or multiple glucose-lowering therapies and

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HYPOGLYCEMIA ASSESSMENT, PREVENTION, AND TREATMENT

Recommendations

1. Review history of hypoglycemia.

Setting and Modifying Glycemic Goals

Glycemic goals and management should be individualized, with focus on avoiding therapeutic inertia to reduce risks of acute and chronic diabetes complications as well as undue treatment burden and adverse effects of therapy. Multiple factors must be considered when setting a glycemic goal, using shared decision-making to integrate individual needs, goals, and preferences and considering characteristics that influence risks and benefits of therapy. In addition to supporting the establishment of individualized glycemic goals, this approach may optimize engagement and self-efficacy.

The factors to consider in individualizing goals are depicted in Fig. 6.1. This figure is not designed to be applied rigidly in the care of a given individual but rather to be used as a framework to guide clinical decision-making and engage people with type 1 and type 2 diabetes in shared decision-making. More aggressive goals may be recommended if they can be achieved safely, with an acceptable burden of therapy (including hypoglycemic risk), and if life expectancy is sufficient to derive benefits of greater glucose lowering. Less stringent goals (e.g., A1C up to 8% [64 mmol/mol]) may be recommended for individuals with complex health status and/or limited life expectancy where benefits of more intensive treatment may not be realized or if the expected risks and burdens of treatment outweigh the potential benefits.

Severe or frequent hypoglycemia is an absolute indication for the modification of treatment plans, including setting higher glycemic goals, if the treatment plan requires use of medications associated with high risk for hypoglycemia (i.e., insulin and insulin secretagogues). Diabetes is a progressive chronic disease; thus, glycemic goals are likely to change over time. It is important to partner with people with diabetes during the deintensification process to understand their goals of diabetes treatment and agree upon appropriate glycemic monitoring, glucose levels, and goals of care.

Hypoglycemia Risk Assessment

1. Screen individuals at risk for hypoglycemia for impaired hypoglycemia awareness at least annually and when clinically appropriate. Refer to a trained health care professional for evidence-based intervention to improve hypoglycemia awareness. 2. Screen individuals at high risk for hypoglycemia or with severe and/or frequent hypoglycemia for fear of hypoglycemia at least annually and when clinically appropriate. Refer to a trained health care professional for evidence-based intervention. 3. Consider an individual's risk for hypoglycemia when selecting diabetes medications and glycemic goals. 4. Use of CGM is beneficial and recommended for individuals at high risk for hypoglycemia.

Treatment of Hypoglycemia

1. Glucose is the preferred treatment for the conscious individual with glucose ≤ 70 mg/dL (≤ 3.9 mmol/L), although any form of carbohydrate that contains glucose may be used. Avoid using foods or beverages high in fat and/or protein for initial treatment of hypoglycemia. Fifteen minutes after initial treatment, repeat the treatment if hypoglycemia persists. 2. Glucagon should be prescribed for all individuals taking insulin or at high risk for hypoglycemia. Family, caregivers, school personnel, and others providing support to these individuals should know its location and be educated on how to administer it. Glucagon preparations that do not have to be reconstituted are preferred. 3. First aid kits should include oral glucose for use in treating hypoglycemia. 4. All individuals taking insulin or at risk for hypoglycemia should receive structured education for hypoglycemia prevention and treatment, with ongoing education for those who experience hypoglycemic events.

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ASSESSMENT

One or more episodes of level 2 or 3 hypoglycemia should prompt reevaluation of the treatment plan, including deintensifying or switching diabetes medications if appropriate. Regularly assess cognitive function; if impaired or declining cognition is found, the clinician, person with diabetes, and caregiver should increase vigilance for hypoglycemia. Hypoglycemia has a broad range of negative health consequences. Level 3 hypoglycemia may be recognized or unrecognized and can progress to loss of consciousness, seizure, coma, or death.

Hypoglycemia Definitions and Event Rates

Hypoglycemia is often the major limiting factor in the glycemic management of type 1 and type 2 diabetes. Recommendations regarding the classification of hypoglycemia are outlined in Table 6.4. Level 1 hypoglycemia is defined as a measurable glucose concentration <70 mg/dL (<3.9 mmol/L) and ≥ 54 mg/dL (≥ 3.0 mmol/L). A blood glucose concentration of 70 mg/dL (3.9 mmol/L) has been recognized as a threshold for adrenergic responses to falling glucose in people without diabetes. Symptoms of hypoglycemia include, but are not limited to, shakiness, irritability, confusion, tachycardia, sweating, and hunger. Because many people with diabetes demonstrate impaired counterregulatory responses to hypoglycemia and/or experience impaired hypoglycemia awareness, a measured glucose level <70 mg/dL (<3.9 mmol/L) is considered clinically important, regardless of symptoms. Level 2 hypoglycemia (defined as a blood glucose concentration <54 mg/dL [<3.0 mmol/L]) is the threshold at which neuroglycopenic symptoms begin to occur and requires immediate action to resolve the hypoglycemic event. If an individual has level 2 hypoglycemia without adrenergic or neuroglycopenic symptoms, they likely have impaired hypoglycemia awareness.

Glycemic criteria/description	Level 1	Level 2	Level 3
Glucose <70 mg/dL (<3.9 mmol/L) and ≥ 54 mg/dL (≥ 3.0 mmol/L)	Glucose <70 mg/dL (<3.9 mmol/L) and ≥ 54 mg/dL (≥ 3.0 mmol/L)	Glucose <54 mg/dL (<3.0 mmol/L)	A severe event characterized by altered mental and/or physical status requiring assistance for treatment of hypoglycemia, irrespective of glucose level

Nevertheless, they reveal a substantial burden of hypoglycemia-related hospital utilization in the community. Level 1 and level 2 hypoglycemia can be ascertained from individual self-report and are strong risk factors for subsequent level 3 hypoglycemia.

Hypoglycemia Risk Assessment

Assessment of an individual's risk for hypoglycemia includes evaluating clinical risk factors as well as relevant social, cultural, and economic factors. Recommendations 6.10–6.20 group individuals with diabetes into two hypoglycemia risk categories with clinical significance. Individuals at risk for hypoglycemia are those treated with insulin, sulfonylureas, or meglitinides; clinically significant hypoglycemia is rare among individuals taking other diabetes medication classes. Individuals at high risk for hypoglycemia are the subset of individuals at risk for hypoglycemia who either have a major hypoglycemia risk factor or have multiple other risk factors (determined by the health care professional incorporating clinical judgment). This risk stratification is based on epidemiologic studies of hypoglycemia risk. Validated tools have been developed to estimate hypoglycemia risk using predominantly electronic health record data. However, these tools do not include all of the important hypoglycemia risk factors, and more research is needed to determine how they can best be incorporated into clinical care.

Among individuals at risk for hypoglycemia, prior hypoglycemic events, especially level 2 or 3 events, are the strongest risk factors for hypoglycemia recurrence. Hypoglycemia history should be assessed at every clinical encounter and should include hypoglycemic event frequency, severity, precipitants, symptoms (or lack thereof), and approach to treatment. It is essential to correlate glucose readings, both from glucose meters and CGM systems, with symptoms and treatment, as individuals may experience and treat hypoglycemic symptoms without checking their glucose level, treat normal glucose values as hypoglycemic, or tolerate hypoglycemia without treatment either because of lack of symptoms or to avoid hyperglycemia.

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Table 6.5—Assessment of hypoglycemia risk among individuals treated with insulin, sulfonylureas, or meglitinides

| Clinical and biological risk factors | Social, cultural, and economic risk factors | | -----

Major risk factors | Major risk factors | | • Recent (within the past 3–6 months) • level 2 or 3 hypoglycemia • Intensive insulin therapy^{*} • Impaired hypoglycemia awareness • Kidney failure • Cognitive impairment or dementia • History of metabolic surgery | • Food insecurity • Low-income status[§] • Housing insecurity • Fasting for religious or cultural reasons • Underinsurance | | Other risk factors | Other risk factors | | • Multiple recent episodes of level 1 hypoglycemia • Basal insulin therapy[^] • Age ≥75 years[†] • Female sex • High glycemic variability[‡] • Polypharmacy • Cardiovascular disease • Chronic kidney disease (eGFR <60 mL/min/1.73 m² or albuminuria) • Neuropathy • Retinopathy • Major depressive disorder • Severe mental illness • Gastroparesis • β-Blocker therapy | • Low health literacy • Alcohol or substance use disorder |

Major risk factors are those that have a consistent, independent association with a high risk for level 2 or 3 hypoglycemia. Other risk factors are those with less consistent evidence or a weaker association. These risk factors are identified through observational analyses and are intended to be used for hypoglycemia risk stratification. Individuals considered at high risk for hypoglycemia are those with ≥1 major risk factor or who have multiple other risk factors (determined by the health care professional incorporating clinical judgment) (88, 89, 94, 96–99, 117, 119, 122, 190, 191, 88, 89, 94, 96–99, 117, 120, 191, 192). Proximal causes of hypoglycemic events (e.g., exercise and sleep) are not included. eGFR, estimated glomerular filtration rate. ^{*}Rates of hypoglycemia are highest for individuals treated with intensive insulin therapy (including multiple daily injections of insulin, continuous subcutaneous insulin infusion, or automated insulin delivery systems), followed by basal insulin, followed by sulfonylureas or meglitinides. Combining treatment with insulin and sulfonylureas further increases hypoglycemia risk. [†]Accounting for treatment plan and diabetes subtype, the oldest individuals (aged ≥75 years) have the highest risk for hypoglycemia in type 2 diabetes; younger individuals with type 1 diabetes are also at very high risk. [‡]Tight glycemic management in randomized trials increases hypoglycemia rates. In observational studies, both low and high A1C are associated with hypoglycemia in a J-shaped relationship. [§]Includes factors associated with low income, such as living in a socioeconomically deprived area.

Individuals at risk for hypoglycemia should also be screened for impaired hypoglycemia awareness (also called hypoglycemia unawareness or hypoglycemia-associated autonomic failure) at least yearly. Impaired hypoglycemia awareness is defined as not experiencing the typical counterregulatory hormone release at low glucose levels or the associated symptoms, which often occurs in individuals with long-standing diabetes or in low-risk individuals, or occur without recurrent hypoglycemia (108). Some individuals with impaired hypoglycemia awareness can sense hypoglycemia, but only at lower glucose levels or with fewer symptoms (109). Individuals with impaired hypoglycemia awareness may experience confusion as the first sign of hypoglycemia, which can create fear of hypoglycemia and severely impact quality of life (109). Impaired hypoglycemia awareness dramatically increases the risk for level 3 hypoglycemia (110). Validated questionnaires for assessing impaired hypoglycemia awareness include the single-question Pedersen-Bjergaard (111) and Gold (112) tools; the Clarke (113) and HypoA-Q (114) tools.

There are a number of important social, cultural, and economic hypoglycemia risk factors that should also be considered. Food insecurity is associated with increased risk of hypoglycemia-related emergency department visits and hospitalizations in low-income households, and this was shown to be mitigated by increased federal nutrition program benefits (121). In general, individuals with low annual household incomes (95), individuals who live in socioeconomically deprived areas (98), and individuals who are underinsured (99) or experiencing housing instability (122) experience higher rates of emergency department visits and hospitalizations for hypoglycemia. Clinicians should also be aware of cultural practices that may influence glycemic management (which are discussed in detail in section 5, “Facilitating Positive Health Behaviors and”).

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Well-being to Improve Health Outcomes”), overestimation of a carbohydrate/meal bolus (134). The acute glycemic response to food correlates better with the glucose content than with the total carbohydrate content. Pure glucose is the preferred initial treatment, but any form of carbohydrate that contains glucose will raise blood glucose. People taking acarbose should treat hypoglycemia with only pure glucose, as acarbose inhibits sucrose breakdown, potentially delaying correction of hypoglycemia if sucrose-containing carbohydrates are used (135). Added fat may slow and then prolong the acute glycemic response. Dietary protein intake may increase insulin secretion and should not be used to treat hypoglycemia (136). Ongoing insulin activity or insulin secretagogues may lead to recurrent hypoglycemia unless more food is ingested after recovery.

Because hypoglycemia may occur in workplaces, schools, and other institutions and public settings (87,91,92,137), first aid kits (both prepackaged and self-made) should include oral glucose for hypoglycemia treatment. Notably, current guidance for first aid kit composition omits glucose (138,139). Options for oral glucose products include glucose tablets, which have the benefits of a long shelf life and easy dosing, and/or glucose gel products, which generally have a shorter shelf life but may be easier to ingest for children and edentulous individuals. Instructions provided with these products should include guidance to only administer oral glucose to an alert person, to check blood glucose before administering oral glucose and 15 min later, and to administer 15 g of glucose as initial treatment.

Hypoglycemia Prevention

A multicomponent hypoglycemia prevention plan (Table 6.7) is critical to caring for individuals at risk for hypoglycemia. Hypoglycemia prevention begins by establishing an individual's hypoglycemia history and risk factors, as discussed in HYPOGLYCEMIA RISK ASSESSMENT above. Structured education for hypoglycemia prevention and treatment is critical and has been shown to improve hypoglycemia outcomes (146,147). Education should ideally be provided through a diabetes self-management education and support program or by a trained diabetes care and education specialist, although these services are not available in many areas (148,149). If structured education is not available, clinicians should educate individuals at risk for hypoglycemia on hypoglycemia definitions, situations that may precipitate hypoglycemia (e.g., fasting, delayed meals, physical activity, and illness), blood glucose self-monitoring, avoidance of driving with hypoglycemia, step-by-step instructions on hypoglycemia treatment as discussed above, and glucagon use as appropriate (146).

Glucagon

The use of glucagon is indicated for the treatment of hypoglycemia in people unable or unwilling to consume carbohydrates by mouth. All individuals treated with insulin or who are at high risk of hypoglycemia as discussed above should be prescribed glucagon. For these individuals, clinicians should routinely review their access to glucagon, as appropriate glucagon prescribing is very low in current practice (140–142). An individual does not need to be a health care professional to safely administer glucagon. Those in close contact with, or having custodial care of, individuals with diabetes prescribed glucagon (e.g., family members, roommates, school personnel, coaches, childcare professionals, correctional institution staff, or coworkers) should be instructed on the use of glucagon, including where the glucagon product is kept and when and how to administer it. It is essential that they be explicitly educated to never administer insulin to individuals experiencing hypoglycemia.

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CGM can be a valuable tool for detecting and preventing hypoglycemia in many individuals with diabetes, and it is recommended for insulin-treated individuals, especially those using multiple daily insulin injections or continuous subcutaneous insulin infusion. There is clinical trial evidence that CGM reduces rates of hypoglycemia in these populations. CGM can reveal asymptomatic hypoglycemia and help identify patterns and precipitants of hypoglycemic events and provide alarms that can warn individuals of falling glucose so that they can intervene (150,151). CGM may also be helpful to reduce hypoglycemia and improve quality of life in people with hypoglycemia as a complication of metabolic surgery (152,153). For more information on using BGM and

CGM for hypoglycemia prevention, see section 7, "Diabetes Technology."

An essential component of hypoglycemia prevention is appropriate modification to diabetes treatment in the setting of intercurrent illness (discussed in detail below) or to prevent recurrent hypoglycemic events. Level 2 or 3 hypoglycemic events in particular should trigger a reevaluation of the individual's diabetes treatment plan, with consideration of deintensification of therapy within individualized glycemic goals.

Individuals with impaired hypoglycemia awareness should be offered training to reestablish awareness of hypoglycemia. Fear of hypoglycemia and impaired hypoglycemia awareness often co-occur, so interventions aimed at treating one often benefit both (154). Several evidence-based training programs have been developed for this purpose and have been demonstrated to reduce rates of hypoglycemia and improve quality of life among people with type 1 diabetes and impaired hypoglycemia awareness (74,155,156). However, these programs are not currently available for clinical use. Similar training can be provided through qualified behavioral health professionals, diabetes care and education specialists, or other professionals with experience in this area, although this approach has not been evaluated in clinical trials. In addition, several weeks of avoidance of hypoglycemia, typically accomplished through a temporary relaxation of glycemic goals, can improve counterregulation and hypoglycemia awareness in many people with diabetes (157). Hence, individuals with impaired hypoglycemia awareness and recurrent hypoglycemic episodes may benefit from short-term relaxation of glycemic goals.

Product	Form	Median AWP* (min, max)	Median NADAC* (min, max)	Dosage
Glucagon	Injection			
powder with diluent for reconstitution		\$303 (\$180, \$337)	\$242 (\$207, \$242)	1 mg
Glucagon	Nasal powder	\$357 \$285 3 mg		
Glucagon	Prefilled pen, prefilled syringe		\$391 \$303 (\$303, \$304)	0.5 mg, 1 mg
Dasiglucagon	Prefilled pen, prefilled syringe	\$371 \$298		0.6 mg

AWP, average wholesale price; max, maximum; min, minimum; NADAC, National Average Drug Acquisition Cost. AWP and NADAC prices are as of 1 July 2025. *Calculated per unit (AWP [192] or NADAC [193]); median AWP or NADAC is listed alone when only one product and/or price is described.

INTERCURRENT ILLNESS

Stressful events (e.g., illness, trauma, and surgery) increase the risk for both hyperglycemia and hypoglycemia among individuals with diabetes, particularly in the setting of underlying clinical complexity and polypharmacy. In severe cases, they may precipitate hyperglycemic crises, which are life-threatening and require immediate medical care. Any individuals with diabetes experiencing illness or other stressful events should be assessed for the need for more frequent monitoring of glucose; ketosis-prone individuals also require urine or blood ketone monitoring. Clinicians should reevaluate diabetes treatment during these events and make adjustments as appropriate. Specifically, clinicians should consider holding metformin and SGLT2 inhibitors when oral intake cannot be maintained or if there is concern for acute kidney injury (158). GLP-1 RAs may need to be held for illnesses with significant gastrointestinal.

Hypoglycemia prevention action	Initial visit	Every follow-up visit	Annual visit
Hypoglycemia history assessment	✓	✓	✓
Hypoglycemia awareness assessment	✓	-	✓
Cognitive function and other hypoglycemia risk factor assessment	✓	-	✓
Structured individual education for hypoglycemia prevention and treatment	✓	✓*	✓*
Consideration of diabetes technology needs	✓	✓	✓
Reevaluation of diabetes treatment plan with deintensification, simplification, or agent modification as appropriate	✓	✓†	✓†
Glucagon prescription and training for close contacts for insulin-treated individuals or those at high hypoglycemic risk	✓	-	✓
Training to improve hypoglycemia awareness	✓‡	-	✓‡

The listed frequencies are the recommended minimum; actions for hypoglycemia prevention should be taken more often as needed based on clinical judgment. *Indicated with recurrent hypoglycemic events or at initiation of medication with a high risk for hypoglycemia. †Indicated with any level 2 or 3 hypoglycemia, intercurrent illness, or initiating interacting medications. ‡Indicated when impaired hypoglycemia awareness is detected.

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Symptoms (158). Thiazolidinediones should be held, and their ultimate reinitiation reassessed on an individualized basis, in the setting of heart failure exacerbation or other conditions with hypervolemia (158). In addition, clinicians should be aware of medication interactions that may precipitate hypoglycemia and modify treatment appropriately. Notably, sulfonylureas interact with a number of commonly used antimicrobials (fluoroquinolones, clarithromycin, sulfamethoxazole-trimethoprim, metronidazole, and fluconazole) that can dramatically increase their effective dose, leading to hypoglycemia (159–161). Clinicians should consider temporarily decreasing or stopping sulfonylureas when these antimicrobials are prescribed. It is important to provide specific, individualized guidance to people with diabetes and their caregivers on steps to prevent, detect, and treat significant hypoglycemia, hyperglycemia, and other adverse events during illness and other stressful events as well as to provide guidance on who to call with questions (including during non-clinical hours) and where to seek help for different levels of clinical acuity with the goal of preventing high-risk illness (158).

For further information on management of hyperglycemia in the hospital, see section 16, “Diabetes Care in the Hospital.”

HYPERGLYCEMIC CRISES: DIAGNOSIS, MANAGEMENT, AND PREVENTION

Recommendations

- 6.21 Review history of hyperglycemic crises (i.e., diabetic ketoacidosis and hyperglycemic hyperosmolar state) at every clinical encounter for all individuals with diabetes at risk for these events. C - 6.22 Provide structured education on the recognition, prevention, and management of hyperglycemic crisis to all individuals with type 1 diabetes, those with type 2 diabetes who have experienced these events, and people at high risk for these events. B

Diabetic ketoacidosis (DKA) and the hyperglycemic hyperosmolar state (HHS) are serious, acute, and life-threatening hyperglycemic emergencies (162). Over the past decade, rates of these hyperglycemic crises have increased in both type 1 and type 2 diabetes (90,163,164). Individuals with type 2 diabetes treated with SGLT2 inhibitors include very-low-carbohydrate diets and prolonged fasting, dehydration, excessive alcohol intake, and the presence of autoimmunity, in addition to typical precipitating factors (173,174). Up to 2% of pregnancies with pregestational diabetes (most often type 1 diabetes) are complicated by DKA. The incidence of DKA in gestational diabetes is low (<0.1%) (175). Pregnant individuals may present with euglycemic DKA (glucose <200 mg/dL [11.1 mmol/L]), and the diagnosis of DKA may be hindered by the presence of mixed acid-based disturbances, particularly in the setting of hyperemesis. Due to significant risk of fetomaternal harm, pregnant individuals at risk for DKA should be counseled on the signs and symptoms suggestive of DKA and seek immediate medical attention if concern for DKA is present.

Hyperglycemic crisis should be considered in all individuals presenting with polyuria, polydipsia, weight loss, vomiting, dehydration, and change in cognitive state. Individuals at risk for DKA should be counseled on the early signs and symptoms of DKA, provided with appropriate tools for ketone measurement (urine and/or blood ketone tests), and educated on timely self-management of hyperglycemia and hyperketonemia (“sick day advice”) (176–178) to prevent clinical deterioration and need for acute care. Clinicians should review DKA prevention practices and availability of ketone monitoring tools at each visit. Individuals treated with intensive insulin therapy should not stop or hold their basal insulin even if not eating, and clinicians should provide detailed instructions on insulin dose adjustments in the setting of illness or fasting to prevent DKA occurrence and worsening. Individuals concerned about or experiencing DKA should be encouraged to contact their diabetes care team immediately. Readily available clinical support can help individuals self-manage hyperglycemia during illness and prevent emergency department and hospital care (179). Individuals at risk for DKA should measure ketones in the presence of symptoms and potential precipitating factors (e.g., illness, missed insulin doses, eating disorders), particularly if glucose levels exceed 200 mg/dL (11.1 mmol/L).

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Table 6.8—Risk factors for hyperglycemic crises

| Type 1 diabetes/absolute insulin deficiency | Risk Factors | |

----- | ----- | | Younger age | | | Prior history of hyperglycemic crises | | | Prior history of hypoglycemic crises | | | Presence of other diabetes complications | | | Presence of other chronic health conditions (particularly in people with type 2 diabetes) | | | Presence of behavioral health conditions (e.g., depression, bipolar disorder, and eating disorders) | | | Alcohol and/or substance use | | | High A1C level | | | Insulin rationing | | | SGLT2 (SGLT1/2) inhibitor use | | | Social determinants of health | |

SGLT, sodium–glucose cotransporter. Data are from McCoy et al. (194), Gibb et al. (195), Randall et al. (196), Thomas et al. (197), and Borden et al. (198).

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- # Glycemic Goals, Hypoglycemia, and Hyperglycemic Crises
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